

江苏百灵农化有限公司

BAILING AGROCHEMICAL CO., LTD

THE FOURTH BRIDGE, JIANGGUAN ROAD JIANGYAN DISTRICT TAIZHOU CITY JIANGSU CHINA

TEL: +86-510-86824607 FAX: +86-510-86823825

MATERIAL SAFETY DATA SHEET

1. Chemical Product Identification

Manufacturer: Bailing Agrochemical Co., Ltd

Add: The Fourth Bridge, Jiangguan Road, Jiangyan District, Taizhou City, Jiangsu, China

Tel: +86-510-86824607 Fax: +86-510-86823825

Product name: LUFENURON 5% W/V EC

2. Composition / Information On Ingredients

Composition	CAS No.	Content (wt, %)
Lufenuron/虱螨脲	103055-07-8	5.4%
Phosphoric acid, octadecyl ester/十八烷基磷酸酯	39471-52-8	5%
Cremophor EL/蓖麻油聚氧乙烯醚	61791-12-6	7%
Castor-oil sulfated sodium salt/蓖麻酸硫酸酯钠盐	8002-33-3	7%
Polyethylene glycol/聚乙二醇	25322-68-3	9%
Epoxidized soya bean oil/环氧大豆油	8013-07-8	30%
Corn oil/玉米油	8001-30-7	36.6%

3. Hazards Identification

Emergency overview: Toxic to crustaceans may cause long-term adverse effects in the aquatic environment.

Health and Environmental: Harmful if inhaled. Can cause mild eye and skin irritation.

Product is a mild skin sensitizer.

Hazardous Decomposition Products: Sensitive to freezing and warm storage conditions. May decompose at high temperatures forming toxic gases.

4. First Aid Measures

Eye: Wash out with plenty of water with the eyelid held wide open for at least 15 minutes.

Immediately get medical attention.

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Skin: Remove contaminated clothing. Wash off with plenty of water and soap. Consult a doctor in the event of any complaints. Wash clothing separately before re-use.

Ingestion: Do not induce vomiting. Wash out mouth with water. Immediately get medical attention.

Inhalation: Remove victim to fresh air. Keep victim warm and at rest. Get medical attention.

NOTE TO PHYSICIAN

There is no specific antidote if this product is ingested. Treat symptomatically.

5. Fire-Fighting Measures

Specific hazards: High volume water jet, (contamination risk), should not be used for safety reason.

Firefighting further advice: Use appropriate extinguishing media for combustibles in the area. Residual organic material will emit toxic fumes when burning. Fire fighters to wear self-contained breathing apparatus if risk of exposure to products of combustion.

Suitable extinguishing media: Water fog (or if unavailable fine water spray), foam, dry agent (carbon dioxide, dry chemical powder).

6. Accidental Release Measures

Eye Protection: Goggles or face-shield

Hand Protection: Chemical-resistant gloves

Body Protection: Heavy duty cotton or synthetic fabric working clothes (e.g. overalls).

Precautionary measures after work: Wash thoroughly (shower, bathe, and wash hair). Change clothing. Thoroughly clean protective gear. Thoroughly clean contaminated equipment with soap or soda solution.

7. Handling and Storage

Handling: Avoid contact with skin, eyes and clothing. Avoid inhalation of fog and vapors. Keep away from sources of ignition. Take precautionary measures against static discharge (transfer operations: check grounding). May form explosive vapor-air mixtures. Do not eat, drink or smoke while working. In addition to the measures usually taken in chemical works like splashproof filling and measuring equipment (including vapour stripping), further personal protection measures may have to be implemented to avoid possible contact with the product.

Storage: Store the product in closed original containers. Protect from light and humidity. Storage depot protected against fire.

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8. Personal Protection

Eye Protection: Goggles or face-shield

Hand Protection: Chemical-resistant gloves

Body Protection: Heavy duty cotton or synthetic fabric working clothes (e.g. overalls).

Precautionary measures after work: Wash thoroughly (shower, bathe, wash hair). Change clothing. Thoroughly clean protective gear. Thoroughly clean contaminated equipment with soap or soda solution.

9. Physical and Chemical Properties

Form: liquid.

Color: Clear, light yellow

Odor: Typical pesticide odour

Specific Gravity/Density: 0.923g/cm³ (20 ° C)

Flash point: 97°C

10. Stability and Reactivity

Stability: Stable under standard conditions.

Conditions to Avoid: Incompatible materials, pesticides with alkaline reaction (lime sulfur, copper).

11. Toxicological Information

Acute Toxicity/Irritation Studies (Finished Product) Ingestion: 11.1

(LD₅₀) / acute oral toxicity

Lufenuron 50g/L EC has to be classified in GHS Category 5 or unclassified for obligatory labeling requirement for oral toxicity, the corresponding LD₅₀ cut-off value being 5000mg/kg body weight.

11.2 (LD₅₀) / acute dermal toxicity

The acute dermal LD₅₀ value of Lufenuron 50g/L EC in Wistar rats was greater than 2000mg/kg body weight. Lufenuron 50g/L EC has to be classified in GHS Category 5 or unclassified for obligatory labeling requirement for dermal toxicity, this category corresponds to an LD₅₀ value lying between > 2000 to ≤ 5000mg/kg body weight.

11.3 (LC₅₀) / acute inhalation

The acute inhalation LC₅₀ value of this product is greater than the maximum attainable concentration of 4.32mg/L and is anticipated to belong either to GHS Category 4 or GHS Category 5.

11.4 eye irritation

This product is classified as category 2 "irritating" to eyes (subcategory 2A) of rabbit.

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11.5 skin irritation

This product is classified as non-irritation to skin of rabbit and could not be classified as per GHS for classification of chemicals.

11.6 sensitization test

This product is classified as non-sensitizer to guinea pig as per the criteria of classification for the Buehler test.

12. Ecological and Ecotoxicological Information

12.1bird

LD₅₀ = >2000 mg/kg (bobwhite quail, mallard ducks)

12.2 fish and aquatic organisms

Fish:

LC₅₀ (96 h) >20 mg/L (Lepomis macrochirus (bluegill))

LC₅₀ (96 h) >73 mg/L (rainbow trout)

LC₅₀ (96 h) >63 mg/L (carp)

Aquatic invertebrates

EC₅₀ (48h) = 0.0072 mg/L (Daphnia magna (water flea)); 48 h

Aquatic plants

Toxicity to aquatic plants:

Growth inhibition, Algae: ErC₅₀ (72 h) = >30 mg/L (Selenastrum capricornutum (green algae))

12.3Honey bee

LC₅₀ (oral) = >197 µg/bee

LC₅₀ (topical) = >200 µg/bee

12.4Non-target organisms

Toxicity to soil dwelling organisms: No adverse effects on earthworms.

13. Disposal Considerations

Disposal: Product Disposal: Pay attention to protective clothing and measures. Cover up product with absorptive material such as sand, soil, diatomaceous earth, etc. Collect material in specially marked, tightly closing containers. Clean dirty areas with water and detergent. Put washing water in containers too, to avoid any contamination of surface and ground water, water supplies and drains. Hose down the area for a prolonged period. Heavily contaminated soil layers have to be dug out down to clean soil. Spilled product cannot be used further and must be disposed of. If safe disposal is not possible, contact the manufacturer, the dealer or the local representative and dispose of in an incinerator approved for chemicals.

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Container Disposal: Dispose of empty containers in an incinerator approved for chemicals. Damaged containers: Place original containers in specially marked larger ones. Check possibilities of recycling large empty containers, drums and barrels.

14. Transport Information

Rail / Road (RID/ADR)

UN-No: 3082

Class: 9

Packaging Group: III

Sea (IMDG-Code)

UN-No: 3082

Class: 9

Packaging Group: III

MARINE POLLUTANT: yes

Air (ICAO/IATA)

UN-No: 3082

Class: 9

Packaging Group: III

15. Regulatory Information

Safety, health and environmental regulations/legislation specific for the substance or mixture
EU

The product is regulated under the EU Directive(s) or Regulation(s) on plant protection products since it is one of plant protection products.

16. Other Information

All information and instructions provided in this Material Safety Data Sheet (MSDS) are based on the current state of scientific and technical knowledge at the date indicated on the present MSDS and are presented in good faith and believed to be correct. This information applies to the product as such. In case of new formulations or mixes, it is necessary to ascertain that a new danger will not appear. It is the responsibility of persons on receipt of this MSDS to ensure that the information contained herein is properly read and understood by all people who may use, handle, dispose or in any way come in contact with the product. If the recipient subsequently produce formulations containing this product, it is the recipients sole responsibility to ensure the transfer of all relevant information from this MSDS to their own MSDS.